

3.3 Differentiation Rules

This section introduces several rules that allow us to differentiate constant functions, power functions, polynomials, rational functions, and certain combinations of them, simply and directly, without having to take limits each time.

① Some Basic Rules and The Power Rule

(a) Derivative of a Constant Function

$$\frac{d}{dx} c = 0, \text{ where } c \text{ is a constant.}$$

(b) Derivative of a Positive Integer Power

$$\frac{d}{dx} x^n = nx^{n-1}, \text{ where } n \text{ is a positive integer.}$$

(c) Power Rule

$$\frac{d}{dx} x^n = nx^{n-1}, \text{ where } n \text{ is any real number.}$$

(d) Derivative Constant Multiple Rule

$$\frac{d}{dx}(cu) = c \frac{du}{dx}, \text{ where } c \text{ is a constant.}$$

Example 1 Differentiate the following powers of x .

$$(a) x^3 \quad (b) x^{\frac{2}{3}} \quad (c) x^{\sqrt{2}} \quad (d) \frac{1}{x^4} \quad (e) x^{-\frac{4}{3}} \quad (f) \sqrt{x^{2+\pi}}$$

Example 2 Differentiate (a) $3x^2$ (b) $-f(x)$.

② Sum Rule

Sun Rule If u and v are differentiable functions of x , then

$$\frac{d}{dx}(u+v) = \frac{d}{dx}u + \frac{d}{dx}v.$$

Example 3 Find the derivative of $y = x^3 + \frac{4}{3}x^2 - 5x + 1$.

Example 4 Does the curve $y = x^4 - 2x^2 + 2$ have any horizontal tangents? If so, where?

③ Product and Quotient Rule

While the derivative of the sum of two functions is the sum of their derivatives, the derivative of the product of two functions is not the product of their derivatives. For instance,

$$\frac{d}{dx}(x \cdot x) = \frac{d}{dx}(x^2) = 2x, \quad \text{while} \quad \frac{d}{dx}(x) \cdot \frac{d}{dx}(x) = 1 \cdot 1 = 1.$$

The derivative of a product of two functions is the sum of two products, as we now explain.

Product and Quotient Rule

If u and v are differentiable functions of x , then

$$\begin{aligned} \frac{d}{dx}(uv) &= u \frac{dv}{dx} + \frac{du}{dx} v. \\ \frac{d}{dx}\left(\frac{u}{v}\right) &= \frac{\frac{du}{dx}v - u \frac{dv}{dx}}{v^2}. \end{aligned}$$

Example 5 Find the derivative of $y = (x^2 + 1)(x^3 + 3)$.

Example 6 Find the derivative of $y = \frac{t^2-1}{t^3+1}$.

Example 7 Find the derivative of $y = \frac{(x-1)(x^2-2x)}{x^4}$.

④ Second- and Higher-Order Derivatives

If $y = f(x)$ is a differentiable function, then its derivative $f'(x)$ is also a function. If f' is also differentiable, then we can differentiate f' to get a new function of x denoted by f'' . So $f'' = (f')'$. The function f'' is called the **second derivative** of f because it is the derivative of the first derivative. It is written in several ways:

$$f''(x) = \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{dy'}{dx} = y'',$$

and

$$y^{(n)} = \frac{dy^{(n-1)}}{dx} = \frac{d^n y}{dx^n}, \quad n \geq 1.$$

Example 8 Find the first four derivatives of $y = x^3 - 3x^2 + 2$.